Antoine Lesage-Landry

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Employment

Department of Electrical Engineering, Polytechnique Montréal
Associate Professor, June 2024 – present

Énergie GridSync Inc.
Chief Scientific Officer, April 2025 – present

Department of Mathematics, Computer Science and Engineering, Université du Québec à Rimouski (UQAR)
Adjunct Professor, November 2024 – present

Department of Electrical Engineering, Nagoya University, Japan
Visiting Associate Professor at the Power Electronics Lab., August 2024 – present

Department of Electrical Engineering, Polytechnique Montréal
Assistant Professor, January 2021 – June 2024

Energy & Resources Group, University of California, Berkeley
Postdoctoral Scholar, August 2019 – December 2020

Dept. of Electrical and Electronic Eng., The University of Melbourne
Visiting Scholar, January 2018 – April 2018 and February 2019

Education

University of Toronto, Toronto, ON, Canada
Ph.D., Electrical & Computer Engineering, November 2019

Polytechnique Montréal, Montréal, QC, Canada
B.Eng., Engineering Physics, June 2015

Professional affiliation

Engineer, Ordre des Ingénieurs du Québec (OIQ)
Member, Group for Research in Decision Analysis (GERAD)
Associate Academic Member, Mila – Québec's AI Institute
Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
IVADO Professor, Institute for Data Valorization (IVADO)
Member, Centre for Systems, Technologies Applications for Radiofrequency and Communications (STARaCom)
Member, Poly-GRAMES Research Center
Member, Réseau québécois de l'énergie intelligente (RQEI)

Publications

Journal Papers (*denotes alphabetic order for mathematics or operations research journals)

J49. Mohamad Aziz, Hussein Suprême, Hanane Dagdougui, and **Antoine Lesage-Landry**. Optimal Energy Trading in Residential Prosumer Clusters via Graphon Mean Field Games. *Electric Power Systems Research*, 263: 113745. February 2027. Presented at the 24th Power Systems Computation Conference (PSCC 2026), June 2026.

J48. Étienne Tremblay and **Antoine Lesage-Landry**. Decentralized Temperature Optimization for discrete-controlled Multi-Zone Commercial Buildings. *Science and Technology for the Built Environment*. September 2026. Under review.

J47. Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Risk-Aware Handover Management Approach for Non-Terrestrial Networks with the NTN-Connect Simulator. *IEEE Transactions on Wireless Communications*. September 2026. Under review.

J46. Yassine Afif, Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Günes Karabulut Kurt. Multi-Agent Reinforcement Learning for Handover Management and Power Allocation in LEO Satellite Networks. *IEEE Journal of Radio Frequency Identification*, 10: 801–816. August 2026.

J45. Louis-Philippe Baillargeon, Maxime Berger, and **Antoine Lesage-Landry**. Joint Mixed-integer Convex Optimization of Path, Speed, and Power in Hybrid-electric Ships. *Transportation Research Part C*. August 2026. Under review.

J44. Vincent St-Pierre, Maxime Berger, and **Antoine Lesage-Landry**. Real-Time Flowmeterless Fuel Consumption Estimation of Marine Diesel Engines. *Applied Ocean Research*. August 2026. Under review.

J43. *Sébastien Le Digabel, **Antoine Lesage-Landry**, Samuel Mendoza, and Christophe Tribes. Parallel Version of the Mesh Adaptive Direct Search Algorithm. *Optimization Letters*. July 2026. Under review.

J42. Marios Saleptsis, **Antoine Lesage-Landry**, Marco Mussetta, and Sonia Leva. Decision-Focused Learning for Optimal Battery Scheduling Under Load and Photovoltaic Power Uncertainty in Microgrids. *Sustainable Energy, Grids and Networks*. July 2026. Under review.

J41. Chao Yin and **Antoine Lesage-Landry**. Convex Training of Lipschitz-regularized Shallow Neural Networks. *Automatica*. May 2026. Under review.

J40. Jean-Luc Lupien, Yuen-Man Pun, Youssef Diouane, Iman Shames, and **Antoine Lesage-Landry**. Online Sketched Newton-Raphson. *Automatica*. May 2026. Under review.

J39. Donald Fisher, Keyhan Sheshyekani, and **Antoine Lesage-Landry**. Distributionally Robust Extended Kalman Filtering for Lithium-ion Battery State-of-Charge Estimation. *IEEE Transactions on Control Systems Technology*. March 2026. Under review.

J38. Yu-Hsin Wu, Philippe Maisonneuve, and **Antoine Lesage-Landry**. Optimal Geospatial Wind Farm and Turbine Placement. *Optimization and Engineering*. March 2026. Under review.

J37. Gautham Reddy, Kürşat Tekbıyık, B. Petersen, **Antoine Lesage-Landry**, Güneş Karabulut Kurt, and İsmail Güvenç. AI-Enabled Wireless Propagation Modeling and Radio Environment Maps for 5G Aerial Wireless Networks. *IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation*, January 2026. Under review.

J36. *Sébastien Le Digabel, **Antoine Lesage-Landry**, Ludovic Salomon, and Christophe Tribes. Efficient search strategies for constrained multiobjective blackbox optimization. *Optimization Methods and Software*. April 2025. Under review.

J35. Victor Darleguy and **Antoine Lesage-Landry**. Hierarchical Constraint Reduction for the Penalized Security-Constrained Optimal Power Flow. *IEEE Control Systems Letters*. June 2026. Accepted. To be presented at the 65th IEEE Conference on Decision and Control (CDC), December 2026.

J34. Feng Li, Ilhan Kocar, and **Antoine Lesage-Landry**. An Optimal Design of Time-of-use Tariff to Mitigate Overloads from Residential Electric Vehicle Charging on Distribution Networks. *IEEE Transactions on Industrial Informatics*, 22 (10): 8685–8695. October 2026.

J33. Kürşat Tekbıyık, Amir Hossein, Fahim Raouf, İsmail Güvenç, Mingzhe Chen, Güneş Karabulut Kurt, and **Antoine Lesage-Landry**. AI-Driven Low-Altitude Economy: Spectrum, Mobility, and Validation. *IEEE Network Magazine*, March 2026. Accepted.

J32. Olivier Bélanger, Jean-Luc Lupien, Olfa Ben Yahia, Stéphane Martel, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Online Convex Optimization for On-Board Routing in High-Throughput Satellites. *Journal of Aerospace Information Systems*, 23(6): 499–506. June 2026.

J31. Feng Li, Ilhan Kocar, and **Antoine Lesage-Landry**. Practical Solutions to Volt-var Optimization under Uncertainty via Blackbox Optimization. *IEEE Transactions on Power Delivery*, 41(3): 1180–1191. June 2026.

J30. Jean-William Lauzon, Ilhan Kocar, and **Antoine Lesage-Landry**. Reliability Assessment of Distribution Systems with Microgrids using Discrete-Time Markov Chains. *IEEE Transactions on Power Delivery*, 41(1): 375–386. February 2026.

J29. Matthias Molénat, Jean Mahseredjian, Nasim Rashidirad, and **Antoine Lesage-Landry**. On the Frequency Variation in Load-flow Calculations for Islanded Alternating Current Microgrids. *Mathematics and Computers in Simulation*, 246: 247–256. August 2026.

J28. Julien Pallage and **Antoine Lesage-Landry**. Wasserstein Distributionally Robust Shallow Convex Neural Networks. *INFORMS Journal on Optimization*, 8(1): 61–93. Winter 2026.

J27. Marc-Antoine Coulombe, Maxime Berger, and **Antoine Lesage-Landry**. Simulation of a Closed-loop dc-dc Converter Using a Physics-informed Neural Network-based Model. *Electric Power System Research*, 252: 112408. January 2026. Presented at the 2025 International Conference on Power Systems Transients (IPST), June 2025.

J26. Amir Ali Farzin, Yuen-Man Pun, Philipp Braun, **Antoine Lesage-Landry**, Youssef Diouane, and Iman Shames. Min-Max Optimisation for Nonconvex-Nonconcave Functions Using a Random Zeroth-Order Extragradient Algorithm. *Transactions on Machine Learning Research*. 2835-2856. September 2025.

J25. Xavier Audet, Kliti Qako, and **Antoine Lesage-Landry**. A Distributionally Robust Optimization Strategy for Virtual Bidding in Two-Settlement Electricity Markets. *Sustainable Energy, Grids and Networks*, 43: 101904. September 2025. Presented at the 12th Bulk Power System Dynamics and Control Symposium (IREP'2025), June 2025.

J24. Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt, Access Inequality in LEO Satellite Networks: A Case Study of High-Latitude Coverage in Northern Québec. *IEEE Open Journal of Vehicular Technology*, 6: 1613–1630. June 2025.

J23. Tao Xue, Mingxuan Zhao, Ilhan Kocar, Mohsen Ghafouri, **Antoine Lesage-Landry**, Siqi Bu, and Ziqing Zhu. Efficient Deep Reinforcement Learning-Based Supplementary Damping Control With a Coordinated RMS Training and EMT Testing Scheme. *IEEE Transactions on Power Delivery*, 40(4): 2300–2313. August 2025.

J22. Jean-Luc Lupien, Iman Shames, and **Antoine Lesage-Landry**. Online Interior-point Methods for Time-varying Equality-constrained Optimization. *IEEE Transactions on Automatic Control*, 70 (4): 2636-2643. April 2025.

J21. Olfa Ben Yahia, Zineb Garroussi, Olivier Bélanger, Brunilde Sansò, Jean-François Frigon, Stéphane Martel, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Evolution of High-Throughput Satellite Systems: A Vision of Programmable Regenerative Payload. *IEEE Communications Surveys and Tutorials*, 27(3): 1565-1597. June 2025.

J20. Olfa Ben Yahia, Zineb Garroussi, Brunilde Sansò, Jean-François Frigon, Stéphane Martel, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. A Scalable Architecture for Future Regenerative Satellite Payloads. *IEEE Wireless Communications Letters*, 14(2): 514-518. February 2025.

J19. Jean-Luc Lupien and **Antoine Lesage-Landry**. Ex Post Conditions for the Exactness of Optimal Power Flow Conic Relaxations. *Electric Power Systems Research*, 238: 111130. January 2025.

J18. Feng Li, Élodie Campeau, Ilhan Kocar, and **Antoine Lesage-Landry**. Inferring Electric Vehicle Charging Patterns from Smart Meter Data for Impact Studies. *Electric Power Systems Research*, 235: 110789. October 2024. Presented at the 23rd Power Systems Computation Conference (PSCC 2024), June 2024.

J17. **Antoine Lesage-Landry** and Julien Pallage. Online Dynamic Submodular Optimization. *Automatica*, 167: 111758. September 2024.

J16. Vincent Mai, Philippe Maisonneuve, Tianyu Zhang, Hadi Nekoei, Liam Paull, and **Antoine Lesage-Landry**. *Multi-agent Reinforcement Learning for Fast-Timescale Demand Response of Residential Loads*, Machine Learning, 113 (8), 5203-5234. August 2024.

J15. Feng Li, Ilhan Kocar, and **Antoine Lesage-Landry**. A Rapid Method for Impact Analysis of Grid-edge Technologies on Power Distribution Networks. *IEEE Transactions on Power Systems*, 39 (1): 1530-1542. January 2024.

J14. **Antoine Lesage-Landry**, Félix Pellerin, Joshua A. Taylor, and Duncan S. Callaway. Optimally Scheduling Public Safety Power Shutoffs. *INFORMS Stochastic Systems*, 13 (4), 438-456. December 2023.

J13. Jean-Luc Lupien and **Antoine Lesage-Landry**. An Online Newton's Method with Linear Time-varying Equality Constraints. *IEEE Control Systems Letters*, 7: 1423-1428. February 2023.

J12. **Antoine Lesage-Landry** and Duncan S. Callaway. Approximated Multi-Agent Fitted Q Iteration. *Systems & Control Letters*, 177: 105563. July 2023.

J11. **Antoine Lesage-Landry** and Duncan S. Callaway. Batch reinforcement learning for network-safe demand response in unknown electric grids. *Electric Power Systems Research*, 212:108375. November 2022. Presented at the 22nd Power Systems Computation Conference (PSCC 2022), June 2022.

J10. **Antoine Lesage-Landry**, Joshua A. Taylor, and Duncan S. Callaway. Online Convex Optimization with Binary Constraints. *IEEE Transactions on Automatic Control*, 66 (12): 6164 - 6170. December 2021.

J9. **Antoine Lesage-Landry**, Joshua A. Taylor, and Iman Shames. Second-order Online Nonconvex Optimization. *IEEE Transactions on Automatic Control*, 66 (10): 4866 - 4872. October 2021.

J8. **Antoine Lesage-Landry**, Han Wang, Iman Shames, Pierluigi Mancarella, and Joshua A. Taylor. Online Convex Optimization of Multi-energy Building-to-grid Ancillary Services. *IEEE Transactions on Control Systems Technology*, 28 (6): 2416 - 2431. November 2020.

J7. **Antoine Lesage-Landry** and Duncan S. Callaway. Dynamic and Distributed Online Convex Optimization for Demand Response of Commercial Buildings. *IEEE Control Systems Letters*, 4 (3): 632-637, July 2020. Presented at the 59th IEEE Conference on Decision and Control (CDC), December 2020.

J6. **Antoine Lesage-Landry**, Siyu Chen, and Joshua A. Taylor. Estimating the Frequency Coupling Matrix from Network Measurements. *IEEE Transactions on Control of Network Systems*, 7 (2): 724 - 733. June 2020.

J5. **Antoine Lesage-Landry**, Iman Shames, and Joshua A. Taylor. Predictive Online Convex Optimization. *Automatica*, 113: 108771, March 2020.

J4. **Antoine Lesage-Landry** and Joshua A. Taylor. A Second-order Cone Model of Transmission Planning with Alternating and Direct Current Lines. *European Journal of Operational Research*, 281 (1): 174-185, February 2020.

J3. Olivier Ouellette, **Antoine Lesage-Landry**, Benjamin Scheffel, Sjœrd Hoogland, F. Pelayo García de Arquer, and Edward H. Sargent. Spatial Collection in Colloidal Quantum Dot Solar Cells. *Advanced Functional Materials*, 3 (1): 1908200. January 2020.

J2. **Antoine Lesage-Landry** and Joshua A. Taylor. Setpoint Tracking with Partially Observed Loads. *IEEE Transactions on Power Systems*, 32 (5): 5615 - 5627, September 2018.

J1. **Antoine Lesage-Landry** and Joshua A. Taylor. The Multi-armed Bandit with Stochastic Plays. *IEEE Transactions on Automatic Control*, 63 (7): 2280-2286, July 2018.

Conference Papers

C39. Yekta Demirci, Kürşat Tekbıyık, Jean-François Frigon, **Antoine Lesage-Landry**, and Güneş Karabulut Kurt. UAV Channel Prediction via Physics-Inspired Kolmogorov-Arnold Networks and Memory. *2027 IEEE Wireless Communications and Networking Conference (WCNC)*. October 2026. Under review.

C38. Maeva Cybelle Zambou-Zoleko, Güneş Karabulut Kurt, and **Antoine Lesage-Landry**. Stochastic Task Mapping for Real-Time Electromagnetic Transient Simulation on Heterogeneous Computing Architectures. *17th ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy 2027)*. October 2026. Under review.

C37. Olfa Ben Yahia, Jean-François Frigon, Brunilde Sansò, Stéphane Martel, Guillaume Mantelet, **Antoine Lesage-Landry**, and Güneş Karabulut Kurt. Chance-Constrained Joint Beam Hopping Scheduling and Bandwidth Allocation for LEO Satellites. *IEEE Middle East Conference on Communications and Networking (MECOM 2026)*. September 2026. Accepted.

C36. M. Ghosh, **Antoine Lesage-Landry**, and Joshua A. Taylor. Copositive Characterizations of Convex Hull Pricing. *62nd Annual Allerton Conference on Communication, Control, and Computing*. September 2026.

C35. Yassine Afif, Ashutosh Balakrishnan, Philippe Martins, Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Güneş Karabulut Kurt. Multi-Agent Reinforcement Learning for Joint Handover Management and Power Allocation in Multi-Orbit Satellite Networks. *2026 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. September 2026. **Best Paper Award.**

C34. Olivier Bélanger, **Antoine Lesage-Landry**, and Sébastien Loranger. Speckle-Based Spectroscopy Using Random Fiber Bragg Grating. *OPTICA Advanced Photonics 2026*. July 2026.

C33. Keijo Jacobs, Renan Morais Furlaneto, Dabeaurard Tho, Nasim Rashidirad, Jean Mahseredjian, and **Antoine Lesage-Landry**. Impedance Model Domains for the Stability Analysis of Multi-Inverter-Based-Resources Systems. *2026 IEEE Power & Energy Society General Meeting*. July 2026.

C32. Dabeaurard Tho, Jean Mahseredjian, **Antoine Lesage-Landry**, Nasim Rashidirad, and Alireza Masoom. Assessment of Stability and Control Interaction Risks in PV Parks with Two-Stage Grid-Forming Control. *2026 IEEE Power & Energy Society General Meeting*. July 2026.

C31. Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Güneş Karabulut Kurt. Handover-Aware Joint Resource Optimization for Power-Efficient LEO Satellite Constellations. *2026 IEEE International Conference on Communications (ICC)*. May 2026.

C30. Kürşat Tekbıyık, Güneş Karabulut Kurt, and **Antoine Lesage-Landry**. PIKAN: Physics-Inspired Kolmogorov-Arnold Networks for Explainable UAV Channel Modelling. *2026 IEEE Aerospace Conference*, March 2026.

C29. Julien Pallage and **Antoine Lesage-Landry**. Sliced-Wasserstein Distance-based Data Selection. *24th International Conference on Machine Learning and Applications*. December 2025.

C28. Théo Laurentin, Patrick Coirault, Emmanuel Moulay, **Antoine Lesage-Landry**, and Jérôme Le Ny. Distributed Combined Space Partitioning and Network Flow Optimization: an Optimal Transport Approach. *64th IEEE Conference on Decision and Control (CDC)*, December 2025.

C27. Yassine Afif, Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Güneş Karabulut Kurt. Joint Satellite Power Consumption and Handover Optimization for LEO Constellations. *2025 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. October 2025.

C26. Samuel M. Muhindo, Hanane Dagdougui, **Antoine Lesage-Landry**, and Hussein Suprême. A Comprehensive Review of Transmission and Distribution Optimal Power Flow Problems for the Integration of Distributed Energy Resources. *IEEE Canada Electrical Power and Energy Conference (EPEC 2025)*. October 2025.

C25. Abraham K. N'Zi, Nasim Rashidirad, Jean Mahseredjian, and **Antoine Lesage-Landry**. Modified and Augmented Nodal Analysis-based Optimal Power Flow. *2025 Annual Conference of the IEEE Industrial Electronics Society (IECON)*, October 2025.

C24. Marie-Christine Paré, Vasken Dermardiros, and **Antoine Lesage-Landry**. Efficient Data-Driven Model Predictive Control for Demand Response of Commercial Buildings. *2025 Annual Conference of the IEEE Industrial Electronics Society (IECON)*, October 2025.

C23. Julie Durette, Gunes Karabulut Kurt, and **Antoine Lesage-Landry**. Task Mapping Strategies for Electric Power System Simulations on Heterogeneous Clusters. *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, September 2025.

C22. Christina G. Soldati, Sébastien Le Digabel, and **Antoine Lesage-Landry**. Blackbox optimization for loss minimization in power distribution networks using feeder reconfiguration. *CIGRE 2025 International Symposium*, September 2025.

C21. Mohamad Charara, Martin de Montigny, Nivine Abou Daher, Hanane Dagdougui, and **Antoine Lesage-Landry**. Mixed-integer Second-Order Cone Programming for Multi-period Scheduling of Flexible AC Transmission System Devices. *CIGRE 2025 International Symposium*, September 2025.

C20. Hannah Merrigan, Wu Yu-Hsin, **Antoine Lesage-Landry**, Koichi Shigematsu, Masayoshi Yamamoto, and Jun Imaoka. Unlocking Free-Position Electric Vehicle Charging: A Neural Network-Driven Approach for Optimisation of Multi-coil Wireless Power Transfer Systems. *Energy Conversion & Congress Expo Europe 2025*. September 2025.

C19. Mohammed Almekhlafi, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Connectivity-Aware Task Offloading for Remote Northern Regions: A Hybrid LEO-MEO Architecture. *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*. September 2025.

C18. Kursat Tekbiyik, Gunes Karabulut Kurt, and **Antoine Lesage-Landry**. Federated Learning for UAV-Based Spectrum Sensing: Enhancing Accuracy Through SNR-Weighted Model Aggregation. *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*. September 2025.

C17. Victor Ricaux, Antoine Legrain, and **Antoine Lesage-Landry**. On the Clique Decomposition Impact to the Optimal Power Flow Semidefinite Relaxation Solve Time. *2025 IEEE Power & Energy Society General Meeting*. July 2025.

C16. Zineb Garroussi, Olfa Ben Yahia, Brunilde Sansò, Jean-François Frigon, Stéphane Martel, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Multi-Priority Scheduling for Traffic Management in Future Scalable Payloads. *IEEE International Conference on Communications: The 5th Workshop on Satellite Mega-Constellations in the 6G Era*. June 2025.

C15. Hannah Merrigan, Wu Yu-Hsin, Koichi Shigematsu, Masayoshi Yamamoto, Jun Imaoka, and **Antoine Lesage-Landry**. Optimising Electric Vehicle Wireless Charging Systems Using Neural Networks to Enable Free-Position Parking. *13th International Conference on Renewable Energy Research and Applications (ICRERA)*. November 2024.

C14. Julien Pallage, Bertrand Scherrer, Salma Naccache, Christophe Bélanger, and **Antoine Lesage-Landry**. Sliced-Wasserstein-based Anomaly Detection and Open Dataset for Localized Critical Peak Rebates. *NeurIPS 2024 Workshop on Tackling Climate Change with Machine Learning*. December 2024.

C13. Jisoo Kim, Jean Mahseredjian, Ulas Karaagac, **Antoine Lesage-Landry**, Ahda P. Grilo Pavani, and Chul-Hwan Kim. State-Space Models for Control Interaction Analysis of DFIG and FSC Wind Parks. *2024 Annual Conference of the IEEE Industrial Electronics Society (IECON)*. November 2024.

C12. Olivier Daoust, Hasan Nayir, Irfan Azam, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Tensor-based Space Debris Detection for Satellite Mega-constellations. *IEEE International Conference on Communications (ICC) 2024 4th Workshop on Satellite Mega-Constellations in the 6G Era*. June 2024.

C11. Matthias Molénat, Nasim Rashidirad, Jean Mahseredjian, and **Antoine Lesage-Landry**. On the frequency variation in load-flow calculations for

islanded AC microgrid. *International Conference on Modeling and Simulation of Electric Machines, Converters and Systems (ELECTRIMACS) 2024*. May 2024.

C10. Olivier Bélanger, Olfa Ben Yahia, Stéphane Martel, **Antoine Lesage-Landry**, and Gunes Karabulut Kurt. Quality of Service-Constrained Online Routing in High Throughput Satellites. *2024 IEEE Aerospace Conference*, March 2024.

C9. Feng Li, Ilhan Kocar, and **Antoine Lesage-Landry**. Mitigating Equipment Overloads due to Electric Vehicle Charging Using Customer Incentives. *2023 IEEE Power & Energy Society General Meeting*, July 2023.

C8. Vincent Mai, Philippe Maisonneuve, Tianyu Zhang, Hadi Nekoei, and **Antoine Lesage-Landry**. Multi-Agent Reinforcement Learning for Fast-Timescale Demand Response of Residential Loads. *22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, June 2023.

C7. Vincent Mai, Philippe Maisonneuve, Tianyu Zhang, José Montalvo, Liam Paull, and **Antoine Lesage-Landry**. Multi-agent Reinforcement Learning for Fast-Timescale Demand Response. *Reinforcement Learning for Real Life Workshop at NeurIPS 2022*, December 2022.

C6. Vincent Mai, Tianyu Zhang, and **Antoine Lesage-Landry**. Multi-agent Reinforcement Learning for Renewable Integration in the Electric Power Grid. *Tackling Climate Change with Machine Learning: Workshop at NeurIPS 2021*, online, December 2021.

C5. Rodrigo Henríquez, **Antoine Lesage-Landry**, Joshua A. Taylor, Daniel Olivares, and Matías Negrete-Pincetic. Managing Load Contract Restrictions with Online Learning. *Signal and Information Processing (GlobalSIP), IEEE Global Conference on*, November 2017.

C4. Amr Mohamed, **Antoine Lesage-Landry** and Joshua A. Taylor. Dispatching Thermostatically Controlled Loads for Frequency Regulation Using Adversarial Multi-armed Bandits. *Electrical Power and Energy Conference (EPEC), 2017 IEEE*, October 2017.

C3. **Antoine Lesage-Landry** and Joshua A. Taylor. Online Convex Optimization for Demand Response. *X Bulk Power Systems Dynamics and Control Symposium, IREP'2017 Symposium.*, August 2017.

C2. **Antoine Lesage-Landry** and Joshua A. Taylor. Learning to Shift Thermostatically Controlled Loads. *Proceedings of the 50th Hawaii International Conference on System Sciences*, January 2017.

C1. Sébastien Loranger, **Antoine Lesage-Landry**, Elton Soares de Lima Filho, Galina Nemova, Noelio O. Dantas, Paulo C. Morais, and Raman Kashyap.

Spectroscopic and life-time measurements of quantum dot doped glass for optical refrigeration: A feasibility study. *SPIE OPTO. International Society for Optics and Photonics*, February 2013.

Seminars and Talks

S32. *Blackbox optimization for electric power distribution systems*, Derivative-Free Optimization Symposium 2026 (DFOS2026), Montréal, Canada. June 2026.

S31. *Online Sketched Newton-Raphson*, SIAM Conference on Optimization 2026 (SIAM OP26), Edinburgh, Scotland, United Kingdom. June 2026.

S30. *Building trust in neural networks for modelling critical systems*, Department of Management Science's Seminar @ University of Strathclyde, Glasgow, Scotland, United Kingdom. June 2026.

S29. *Building trust in neural networks for modelling in engineering*, CÉDRIC Seminar @ Conservatoire national des arts et métiers (CNAM), Paris, France. May 2026.

S28. *Building trust in neural networks for energy modelling*, Talk@Simlab, Politecnico di Milano, Milan, Italy. January 2026.

S27. *Introduction to Optimization and Machine Learning*, Nagoya University Institute of Materials and Systems for Sustainability's Power Electronic Lab, Nagoya, Japan. September 2025.

S26. *Building trust in neural networks for energy modelling*, Institute of Materials and Systems for Sustainability's Power Electronic Lab, Nagoya, Japan. September 2025.

S25. *Building trust in neural networks for energy modelling*, Symposium "Digital Intelligence for Climate Change Mitigation," Montréal, QC, Canada. March 2025.

S24. *Online Optimization for Real-time and Safe Decision-Making*, Research & Development Defense Canada Seminar, Valcartier, QC, Canada. January 2025.

S23. *Power distribution system analysis: electric vehicle charging detection & integrated volt-var optimization*, Toronto Hydro Seminar, Toronto, ON, Canada (online). January 2025.

S22. *Building trust in neural networks for energy modelling.* Energy Modelling Hub Annual Forum 2024, Ottawa, Canada. December 2024.

S21. *Learning for Electric Power Systems Operations and Planning*, Mila Partner Symposium Fall 2024, Montréal, QC, Canada. October 2024.

S20. *Online Interior-point Methods for Real-time Optimal Power Flow*, RC Control Network @ Hitachi Energy. Montréal, QC, Canada (online). October 2024.

S19. *Introduction to Power Systems Optimization*, Haiti Summer and Fall School 2024. Port-au-Prince, Haiti (online). September 2024.

S18. *Analyzing and Mitigating Impacts of Grid-edge Technologies on Power Distribution Networks*, Toronto Hydro Seminar. Toronto, ON (online). January 2024.

S17. *Analyzing and Mitigating Impacts of Grid-edge Technologies on Power Distribution Networks*, Energy Modelling Hub 2023 Annual Forum. December 2023.

S16. *Analyzing and Mitigating Impacts of Grid-edge Technologies on Power Distribution Networks*, University of Arizona: Department of Mathematics Seminar, Tucson, AZ, USA. October 2023.

S15. *Online Decision-Making*, Northwestern University: ELEC_ENG 495 Special Topics in Electrical Engineering – Introduction to Smart Grid Systems, Evanston, IL, USA. October 2023.

S14. *Online Decision-Making*, The Hong Kong Polytechnique University: Department of Electrical and Electronic Seminar, Hong Kong SAR, China. August 2023.

S13. *Analyzing and Mitigating Impacts of Grid-edge Technologies on Power Distribution Networks*, The Hong Kong Polytechnique University: Department of Electrical and Electronic Seminar, Hong Kong SAR, China. August 2023.

S12. *Online Decision-Making*, Denmark Technical University (DTU) PES Summer School, Copenhagen, Denmark. June 2023.

S11. *Analyzing and Mitigating Impacts of Grid-edge Technologies on Power Distribution Networks*, University of Toronto: ECE Seminar, Toronto, ON, Canada. February 2023.

S10. *Optimally Scheduling Public Safety Power Shutoffs*, University of Michigan: Michigan Power and Energy Laboratory Seminar, Ann Arbor, MI, USA. February 2023.

S9. *A Rapid Method for Impact Analysis of Grid-edge Technologies on Power Distribution Networks*, Resilient Electric Grid Consortium of North America Symposium 2022 (RECONS 2022), College Station, Texas, USA. November 2022.

S8. *Optimally Scheduling Public Safety Power Shutoffs*, University of British Columbia: ECE Seminar, Vancouver, Canada. June 2022.

S7. *Optimally Scheduling Public Safety Power Shutoffs*, 2022 CORS/INFORMS International Meeting, Vancouver, Canada. June 2022.

S6. *Optimally Scheduling Public Safety Power Shutoffs*, Northeastern University: MIE Seminar Series, Boston, United States. May 2022.

S5. *Batch Reinforcement Learning for Network-Safe Demand Response.* ETH Zürich: Power Systems Laboratory Seminar, Zürich, Switzerland/online. March 2021.

S4. *Real-time Decision-Making for Demand Response.* University of California, San Diego: Energy Seminar, San Diego, CA/online. November 2021.

S3. *Real-Time Decision-making for Demand Response Under Uncertainty.* GERAD Webinar. Montréal, QC/online, March 2021.

S2. *Renewable Integration & Demand Response.* ECE1476 – LEDs & Solar Cells, University of Toronto, Toronto, ON. November 2018.

S1. *Online Learning for Demand Response.* The University of Melbourne, Australia: Electrical & Electronic Engineering Seminar, Melbourne, VIC, Autralia. February 2018.

Teaching

Department of Electrical Engineering, Polytechnique Montréal

- ELE2700 – Analyse des signaux (Signal & Systems), Fall 2021–2024, Fall 2026; *coordinator*: Summer 2024–2025, Fall 2025, Fall 2026
- ELE8453 – Méthodes d'optimisation et d'apprentissage pour les réseaux électriques (Optimisation & Learning Methods for Power Systems), Winter 2022–2025; *coordinator*: Winter 2026;
- ELE8456 – Réseaux de distribution (Distribution Networks), *coordinator*, Winter 2022–2024;
- ELE8452 – Réseaux électriques (Power System Analysis), *coordinator*, Fall 2022 and 2023;
- ELE8459 – Protection des réseaux électriques (Power System Proctections), *coordinator*, Winter 2023 and 2024

Supervision

Graduate students & fellows (Electrical Engineering)

Ph.D.

PhD-15. Étienne Tremblay, 2026 – present;

PhD-14. Hannah Merrigan, 2026 – present;

PhD-13. Vincent St-Pierre, 2025 – present (at University of Sherbrooke, co-supervised with Prof. Maxime Berger);

PhD-12. Christina G. Soldati, 2025 – present (co-supervised with Prof. Sébastien Le Digabel);

PhD-11. Lanyue Tang, 2025 – present (Applied Mathematics, co-supervised with Prof. Antoine Legrain);

PhD-10. Chao Yin, 2025 – present;

PhD-9. Dabeaurard Tho, 2024 – present (co-supervised with Prof. Jean Mahseredjian);

PhD-8. Yi Chen, 2024 – present (co-supervised with Prof. Gunes Karabulut Kurt);

PhD-7. Marc-Antoine Coulombe, 2024 – present (at University of Sherbrooke, co-supervised with Prof. Maxime Berger);
PhD-6. Adam Osmani, 2024 - present (co-supervised with Prof. Karabulut Kurt);
PhD-5. Mahmoud Elkahawy, 2024 – present (co-supervised with Prof. Keyhan Sheshyekani);
PhD-4. Ulrich Ephraim Yepmou-Kepnang, 2024 – present (co-supervised with Prof. Jean Mahseredjian);
PhD-3. Théo Laurentin, 2023 – present (co-supervised with Prof. Jérôme Le Ny);
PhD-2. Abraham K. N'Zi, 2023 – present (co-supervised with Prof. Jean Mahseredjian);
PhD-1. Feng Li, 2021 – 2024 (co-supervised with Prof. Ilhan Kocar).

M.A.Sc.

MASc-28. David Bouthot, 2026 – present;
MASc-27. Félix Santerre, 2026 – present (UQAR, co-supervised with Prof. Mohammed Bahoura and Prof. Maxime Berger);
MASc-26. Olivier Bélanger, 2025 – present (co-supervised with Prof. Sébastien Loranger);
MASc-25. Laure-Anne Margaux Réau, 2025 – present (co-supervised with Prof. Audrey Bazerghi);
MASc-24. Louis-Émile Lalonde, 2025 – present;
MASc-23. Tudor-Mihai Buḫut, 2025 – present (co-supervised with Prof. Sébastien Le Digabel);
MASc-22. Julien Courcoul, 2024 – present (co-supervised with Prof. Jean Mahseredjian);
MASc-21. William Zhang, 2024 – present;
MASc-20. Louis-Philippe Baillargeon, 2024 – 2026 (co-supervised with Prof. Maxime Berger);
MASc-19. Mohamad Charara, 2023 – 2026 (Applied Mathematics, co-supervised with Prof. Hanane Dagdougui);
MASc-18. Jérôme Lafontaine, 2024 – 2026 (co-supervised with Prof. Maxime Berger);
MASc-17. Yassine Afif, 2025 – 2026 (co-supervised with Prof. Karabulut Kurt);
MASc-16. Donald Fisher, 2024 – 2026 (co-supervised with Prof. Keyhan Sheshyekani);
MASc-15. Xavier Audet, 2024 – 2025 (Energy Engineering);
MASc-14. Vincent St-Pierre, 2023 – 2025 (at UQAR, co-supervised with Prof. Maxime Berger from 2024);
MASc-13. Victor Darleguy, 2023 – 2025 (Energy Engineering);
MASc-12. Julie Durette, 2023 – 2025 (co-supervised with Prof. Gunes Karabulut Kurt);
MASc-11. Julien Pallage, 2023 – 2025 (Polytechnique Montréal Best Master's Thesis Award, April 2026);
MASc-10. Étienne Tremblay, 2023 – 2025;
MASc-9. Christina G. Soldati, 2023 – 2025 (co-supervised with Prof. Sébastien Le Digabel);
MASc-8. Samuel Mendoza, 2023 – 2024 (co-supervised with Prof. Sébastien

Le Digabel);

MASc-7. Matthias Molénat, 2022 – 2024 (Energy Engineering, co-supervised with Prof. Jean Mahseredjian);

MASc-6. Olivier Bélanger, 2023 – 2024 (co-supervised with Prof. Gunes Karabulut Kurt);

MASc-5. Fatemeh Rajabi, 2022 – 2024 (Applied Mathematics, co-supervised with Prof. Antoine Legrain);

MASc-4. Marie-Christine Paré, 2021 – 2023 (Energy Engineering);

MASc-3. Jean-William Lauzon, 2022 – 2023 (co-supervised with Prof. Ilhan Kocar);

MASc-2. Philippe Maisonneuve, 2022 – 2023 (Energy Engineering);

MASc-1. Jean-Luc Lupien, 2022 – 2023.

M.Eng.

MEng-22. Guillaume Robin, 2026 – present (Energy Engineering);

MEng-21. Oscar Fajardo Montalvo, 2026 – present (Energy Engineering);

MEng-20. Thiery Huber, 2026 – present (Energy Engineering);

MEng-19. Nirmine Douhabi, 2026 – present (Energy Engineering);

MEng-18. Domitille Stubler, 2026 – present (Energy Engineering);

MEng-17. Lazare Bousquet, 2026 – present (Energy Engineering);

MEng-16. Zoé Coirre, 2026 – present (Energy Engineering);

MEng-15. Célian Aragon, 2026 – present (Energy Engineering);

MEng-14. Kimberley de la O, 2026 – present (Energy Engineering);

MEng-13. Anh Tu Nguyen, 2025 – present (Energy Engineering);

MEng-12. Aliou Gaye, 2025 – present;

MEng-11. Romain Jonon, 2025 – present;

MEng-10. Louis-Philippe Parent, 2023 – present (Energy Engineering);

MEng-9. Laurence Leblanc, 2023 – present (Energy Engineering);

MEng-8. Lotfi Labassi, 2025 – 2026 (Energy Engineering);

MEng-7. Baptiste Salvi, 2024 – 2025 (Energy Engineering);

MEng-6. Raphaël Grenier, 2023 – 2023 (Energy Engineering);

MEng-5. Élodie Campeau, 2022 – 2023 (Energy Engineering);

MEng-4. Laurella Dionisi, 2022 – 2023 (Energy Engineering);

MEng-3. Ulrich Ephraim Yepmou-Kepnang, 2022 – 2023 (Energy Engineering);

MEng-2. Anne-Marie Sauvageau, 2021 – 2023 (Energy Engineering);

MEng-1. Inès Conde, 2022 – 2022 (Energy Engineering).

Interns

I-10. Jia Fournier (B.Eng.), 2026 – present;

I-9. Léon Biner (B.Eng.), Summer 2025, 2025 – 2026 (co-supervised with Prof. Sébastien Le Digabel);

I-8. Jérôme Lafontaine (B.Eng.), Summer 2024;

I-7. Hugo Léveillé (B.Eng.), Summer 2024 (co-supervised with Prof. Sébastien Loranger);

I-6. Tudor-Mihai Buḥut (B.Eng.), Summer 2024, 2024 – 2025 (co-supervised with Prof. Sébastien Le Digabel);

I-5. Adam Osmani (B.Eng.), Summer 2023, 2023 - 2024 (co-supervised with Prof. Karabulut Kurt);
I-4. Victor Darleguy (B.Eng.), Summer 2023;
I-3. Julien Pallage (B.Eng.), Summers 2022 & 2023;
I-2. Olivier Daoust (B.Eng.), Summer 2022, 2022 – 2023 (co-supervised with Prof. Karabulut Kurt);
I-1. Félix Pellerin (B.Eng.), Summer 2021.

Postdoctoral fellows

PDF-11. Mohamad Aziz, 2026 – present;
PDF-10. Noureddine Toumi, 2025 – present;
PDF-9. Kürşat Tekbıyık, 2025 – present (co-supervised with Prof. Gunes Karabulut Kurt);
PDF-8. Mohamad Aziz, 2025 – 2026 (co-supervised with Prof. Hanane Dagdougui);
PDF-7. Mohammed Almekhlafi, 2024 – present (co-supervised with Prof. Gunes Karabulut Kurt);
PDF-6. Olfa Ben Yahia, 2023 – present (Mathematics, co-supervised with Prof. Gunes Karabulut Kurt, Brunilde Sanso, Jean-François Frigon);
PDF-5. Noureddine Toumi, 2024 – 2025 (co-supervised with Prof. Jean Mahseredjian);
PDF-4. Ludovic Salomon, 2024 – 2025 (co-supervised with Prof. Sébastien Le Digabel);
PDF-3. Samuel Muhindo, 2024 – 2024 (Mathematics, co-supervised with Prof. Hanane Dagdougui);
PDF-2. Zineb Garroussi, 2023 – 2024 (co-supervised with Profs. Gunes Karabulut Kurt, Brunilde Sanso, Jean-François Frigon);
PDF-1. Christian Bingane, 2022 – 2023 (Mathematics, co-supervised with Prof. Hanane Dagdougui).

Research Associate

RA-2. Victor Darleguy, 2026 – present (part-time);
RA-1. Étienne Tremblay, 2025 – 2026.

Visitors

V-11. Theodoros Xanakis (M.Sc., Norwegian Institute of Technology, NTNU), Summer 2026;
V-10. Marios Saleptsis (Ph.D., Politecnico di Milano, Italy), Fall 2025;
V-9. Baptiste Zoghby (B.Eng., École Polytechnique), Summer 2025 (co-supervised with Prof. Antoine Legrain);
V-8. Hannah Claire Merrigan (M.Sc., Nagoya University, Japan), Winter 2025;
V-7. William Nzitat Yomi (M.Sc., University of Mons, Belgium), Winter 2025;
V-6. Federico Rossi (Ph.D., Politecnico di Milano, Italy), Fall 2024;
V-5. Victor Ricaux (B.Eng., École Polytechnique), Summer 2024 (co-supervised with Prof. Antoine Legrain);
V-4. Joséphine Gobert (M2, ENAC, France), Summer 2024 (co-supervised with Prof. Sébastien Le Digabel);

V-3. Yu-Hsin (Larry) Wu (Ph.D., Nagoya University, Japan), Summer 2023;
V-2. Alexis Caraud (M2, ENAC, France), Summer 2023 (co-supervised with Prof. Sébastien Le Digabel);
V-1. Stéphane Salim (M2, ENAC, France), Summer 2023 (co-supervised with Prof. Sébastien Le Digabel).

Funding

Active grants (PI: principal investigator; co-PI: co-principal investigator)

GA-17. *AuTonomous and resiLient connectivity for sAtellite networkS (ATLAS)* ($1,560,000). NSERC Alliance, Mitacs Accelerate with Terrestar. September 2026 – August 2031. co-PI (PI: G. Karabulut Kurt);

GA-16. *Advancing performance and functional capabilities in next-generation electric powersports vehicles* ($180,000). Mitacs Accelerate with BRP. October 2026 – October 2028. co-PI (PI: M. Berger);

GA-15. *Safe and deployable reinforcement learning methods for voltage control in renewable power systems* ($562,170). NSERC Alliance Advantage with Hydro-Québec. September 2026 – August 2031. **PI**;

GA-14. *Competition and risk-aware optimal bidding in wholesale electricity markets via mean-field control* ($150,000). Mitacs Accelerate with Hydro-Québec. June 2026 – June 2028. **PI**;

GA-13. *Development of a transversal and modular artificial intelligence model for energy system asset management* ($90,000). Mitacs Accelerate with Hydro-Québec. January 2026 – December 2028. **PI**;

GA-12. *Measurement-based modeling and stability analysis of inverter-based resources (IBRs) using machine learning* ($250,833). Mitacs Accelerate with Hydro-Québec. June 2025 – June 2029. co-PI (PI: J. Mahseredjian);

GA-11. *Advanced algorithms for power system planning and operation with non-synchronous energy sources* ($391,303). NSERC Alliance Advantage with Hydro-Québec. May 2025 – May 2030. **PI**;

GA-10. *Learning optimization proxies for sequential decision-making problems: application to power systems* ($225,000). FRQNT-NSERC NOVA. May 2025 – April 2028. co-PI (PI: A. Legrain);

GA-9. *Modernization of modelling and simulation techniques for evaluating protection and stability in inverter-based networks* ($150,000). FRQNT Team Research Project. May 2025 – April 2028. co-PI (PI: I. Kocar);

GA-8. *Towards the electrification of commercial fishing vessels in Quebec: stakeholder mobilization and energy consumption optimization* ($300,000). FRQNT Partnership Research Program in the Maritime Sector. November 2024 – November 2027. co-PI (PI: M. Berger);

GA-7. *Collaborative Research: NSF-NSERC: Pricing in discrete markets using copositive duality* ($549,000). NSERC, NSF (NSERC–NSF Collaboration on Quantum Science and Artificial Intelligence). September 2024 – August 2027. **PI**;

GA-6. *Machine learning-based modelling and optimization for multi-coil transmitter design of electric vehicle wireless charging systems* ($25,000). NSERC Alliance International Catalyst with Nagoya University, Japan. September 2024 – September 2027. **PI**;

GA-5. *Artificial intelligence enabled harmonious wireless coexistence for 3D networks (3D-HARMONY)* ($662,202). NSERC, NSF (NSERC–NSF Collaboration on Quantum Science and Artificial Intelligence). August 2024 – August 2027. co-PI (PI: G. Karabulut Kurt);

GA-4. *Machine learning-accelerated power electronics modelling for real-time simu-*

lations ($690,000). NSERC Alliance-Accelerate, Mitacs. June 2024 – June 2029. **PI**;

GA-3. *Design optimization for hybrid fishing ships* ($280,000). Ministère de l'Environnement, de la Lutte contre les changements climatiques, de la Faune et des Parcs. June 2024 – May 2028. co-PI (PI: A. Boudreau);

GA-2. *Traffic and propagation aware beam management for high throughput satellites* ($584,134). NSERC CRIAQ-Alliance, CRIAQ with MDA Space. June 2024 – May 2027. co-PI (PI: G. Karabulut Kurt);

GA-1. *Real-time optimization for renewable electric power systems* ($227,500). NSERC Discovery Grant and Discovery Launch Supplement. April 2023 – March 2028. **PI**.

Completed grants

GC-9. *Optimization of future energy systems* ($2,070,000). Mitacs, NSERC Alliance with Hydro-Québec. June 2022 – June 2026. co-PI (PI: H. Dagdougui);

GC-8. *Distributionally robust optimal virtual bidding for two-settlement electricity markets* ($53,333). Mitacs. January 2024 – December 2025. **PI**;

GC-7. *A hybrid learning and distributed optimization method for multi-zone building control* ($53,333). Mitacs Accelerate Fellowship with BrainBox AI. May 2023 – August 2025. **PI**;

GC-6. *Tensor-based task mapping optimization for real-time power system simulations on heterogeneous clusters* ($53,333). Mitacs Accelerate Fellowship with OPAL-RT Technologies. September 2023 – August 2025. **PI**;

GC-5. *Trustworthy machine learning for Quebec's virtual power plant* ($30,000). Mitacs Accelerate Rapid Short-term Funding with Hydro-Québec's Hilo. December 2023 – December 2024. **PI**;

GC-4. *Intelligent load balancing for satellite networks (ICARUS)* ($233,650). Mitacs, CRIAQ with MDA Space. December 2022 – March 2024. co-PI (PI: G. Karabulut Kurt);

GC-3. *Optimal building community operation using data-driven mixed-integer convex model predictive control* ($30,000). Mitacs Accelerate Rapid Short-term Funding with BrainBox AI. May 2022 – April 2023. **PI**;

GC-2. *IVADO professor operating fund* ($50,000). IVADO. January 2021 – December 2022. **PI**;

GC-1. *IVADO professor start-up fund* ($30,000). IVADO. January 2021 – December 2022. **PI**.

Service

Journal referee
Automatica, European Journal of Operational Research, IEEE Transactions on {Automatic Control, Control on Network Systems, Control Systems Technology, Power Systems, Smart Grid}, IEEE Control Systems Letter, IEEE Journal of Selected Topics in Signal Processing
Conference referee
IEEE-PES General Meeting, IEEE Conference on Decision and Control, Power Systems Computation Conference
Event organization
General Chair, LORER Summer School 2027 (August 2027).